

Assignment-5

RAVI

1) In the question, I have been requested to evaluate whether customer relationship management affects the financial position of the firm. So, here two hypotheses will be tested one against the other. We have either CRM related to performance or CRM is not related.

We all know that customer relationship management is related to the firm's financial position. so, I choose it as H_0 as it is true and I question H_0 by using H_a which is CRM not related (not true) to test whether H_0 is true or not.

2) According to the experiment, new packages are kept in the big store for sale and the regular packing cookies are kept in other stores. On the comparison of the sales of two packages, we have to decide the hypothesis.

As a firm owner, I choose H_0 as old packaging kept in other stores and I question H_0 by using new packaging products kept in big stores i.e., H_a whether it is true or not. By this relation, we can conclude whether new packing is effective or not.

3) Given sample $p=\{\text{yes,no,yes,yes,yes,no}\}$

$X_i=1$ if yes, 0 if no

Then we get the equation $= (p)^4 \cdot (1-p)^2$

When $p=0.2 \Rightarrow 0.001024$, $p=0.8 \Rightarrow 0.016384$

$p=0.3 \Rightarrow 0.003969$, $p=0.9 \Rightarrow 0.006561$

The proportion of Yes given in the observed value is the highest value in the graph and is called likely hood function $= \text{no of yes} / \text{no of yes+no}$

$\Rightarrow 2/3 \Rightarrow 0.0219478$.

4) Given sample $p=\{\text{yes,no,no,yes,no,no}\}$

$X_i = 1$ if yes, 0 if no

Then we get the equation $= (p)^2 \cdot (1-p)^4$

When $p=0.2 \Rightarrow 0.016384$, $p=0.8 \Rightarrow 0.001024$

$p=0.3 \Rightarrow 0.021609$, $p=0.9 \Rightarrow 0.000081$

The proportion of Yes given in the observed value is the highest value in the graph and is called likely hood function = no of yes / no of yes + no

$\Rightarrow 1/3 \Rightarrow 0.021944$.

5) According to the given model when we change the sample size then it results in a change in the curve(or) maximum likelihood value explained by the value of the probability of the sample.

6) Given,

$H_0 : p=0.4$

$H_a : p=0.6$

Rejection rule = $L(0.4)/L(0.6) < K$, $X_i = 5/2 \Rightarrow 2.5$

For the first type of error $K=1$, the probability of chance of a mistake when my sample likely comes from H_a but it comes from H_0 is 32%.

#Problem 2.1

#2.1.1) given predictor- height and response-weight.

```
library(alr4)
```

```
## Loading required package: car
```

```
## Loading required package: Chordata
```

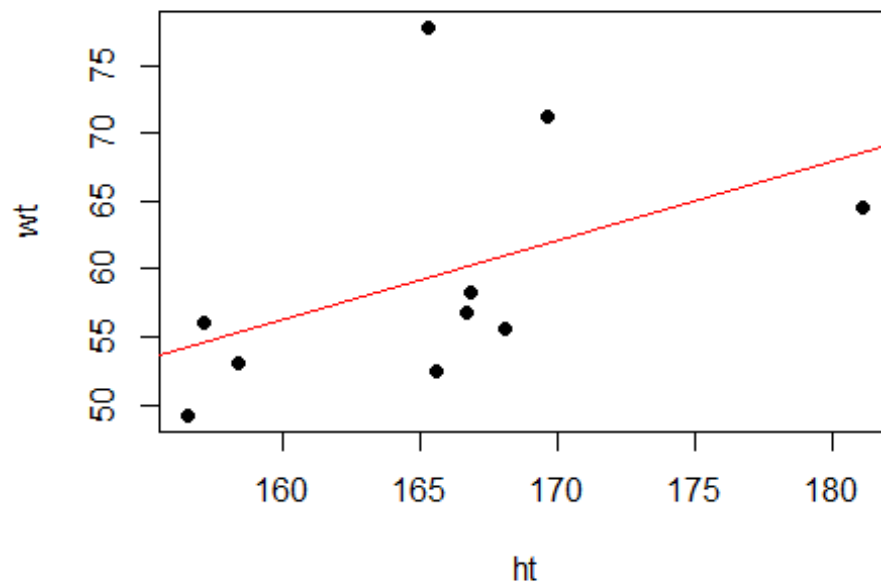
```
## Loading required package: effects
```

```
## lattice theme set by effectsTheme()
```

```
## See ?effectsTheme for details.
```

```
with(Httwt, plot(wt~ht, pch=19))
```

```
with(Httwt, abline(lm(wt~ht), col="red"))
```



*#(it is very hard to say about linearity using 10 observations
 # but it may be possible because of the straight line)*

```
m1 <-lm(wt~ht,Htwt)
summary(m1)
```

```
##
## Call:
## lm(formula = wt ~ ht, data = Htwt)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.1166 -4.7744 -2.8412  0.5696 18.4581
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -36.8759    64.4728  -0.572   0.583
## ht           0.5821     0.3892   1.496   0.173
##
## Residual standard error: 8.456 on 8 degrees of freedom
## Multiple R-squared:  0.2185, Adjusted R-squared:  0.1208
## F-statistic: 2.237 on 1 and 8 DF,  p-value: 0.1731
```

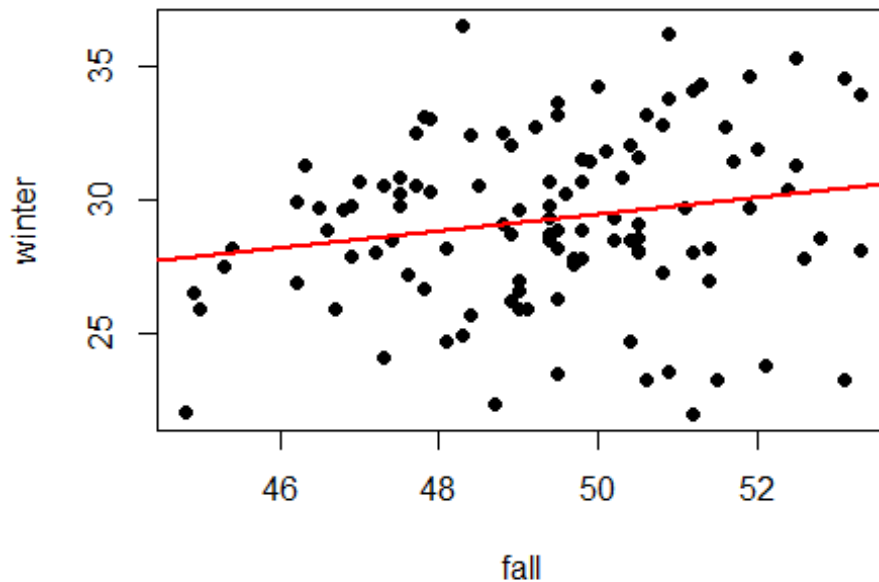
#problem 2.6

```
library(alr4)
data= ftcollinstemp
```

#2.6.1

#plotting the average fall temperature and the average winter temperature

```
with(ftcollinstemp, plot(winter~fall,pch=19))
with(ftcollinstemp, abline(lm(winter~fall),col="red",lwd=2))
```



```
m2<-lm(winter~fall,ftcollinstemp)
summary(m2)

##
## Call:
## lm(formula = winter ~ fall, data = ftcollinstemp)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -7.8186 -1.7837 -0.0873  2.1300  7.5896
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  13.7843     7.5549   1.825  0.0708 .
## fall         0.3132     0.1528   2.049  0.0428 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.179 on 109 degrees of freedom
## Multiple R-squared:  0.0371, Adjusted R-squared:  0.02826
## F-statistic:  4.2 on 1 and 109 DF,  p-value: 0.04284
```

*#there is no linearity in the model and the variance is also not
constant multiple r2 values are also too low so there is no relation.*

#2.6.2

#Test of hypothesis for slope

0.3132/0.1528

```
## [1] 2.049738
```

```
qt(1-0.1/2 , 109)
```

```
## [1] 1.658953
```

*# we will reject the null hypothesis because the t value is more
the quartile value.*

#2.6.3

*# In the summary the multiple R^2 value is 0.0371 twhichimplies that
3.7% of the variability of average winter temp is explained by average fall*

.

#2.6.4

#dividing the data into 2 time periods

from 1900 to 1989

```
m00<-data[1:90,]
```

```
m3 <-lm(winter~fall,m00)
```

```
summary(m3)
```

```
##
```

```
## Call:
```

```
## lm(formula = winter ~ fall, data = m00)
```

```
##
```

```
## Residuals:
```

```
##      Min       1Q   Median       3Q      Max
```

```
## -6.8976 -1.6349  0.0118  2.0079  7.3387
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept)  22.7079      8.2600   2.749  0.00725 **
```

```
## fall         0.1209      0.1681   0.719  0.47397
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## Residual standard error: 3.057 on 88 degrees of freedom
```

```
## Multiple R-squared:  0.005842,    Adjusted R-squared:  -0.005455
```

```
## F-statistic: 0.5171 on 1 and 88 DF,  p-value: 0.474
```

from 1990 to 2010.

```
m90<-data[90:length(data$year),]
```

```
m4<-lm(winter~fall,m90)
```

```
summary(m4)
```

```
##
```

```
## Call:
```

```
## lm(formula = winter ~ fall, data = m90)
```

```
##
```

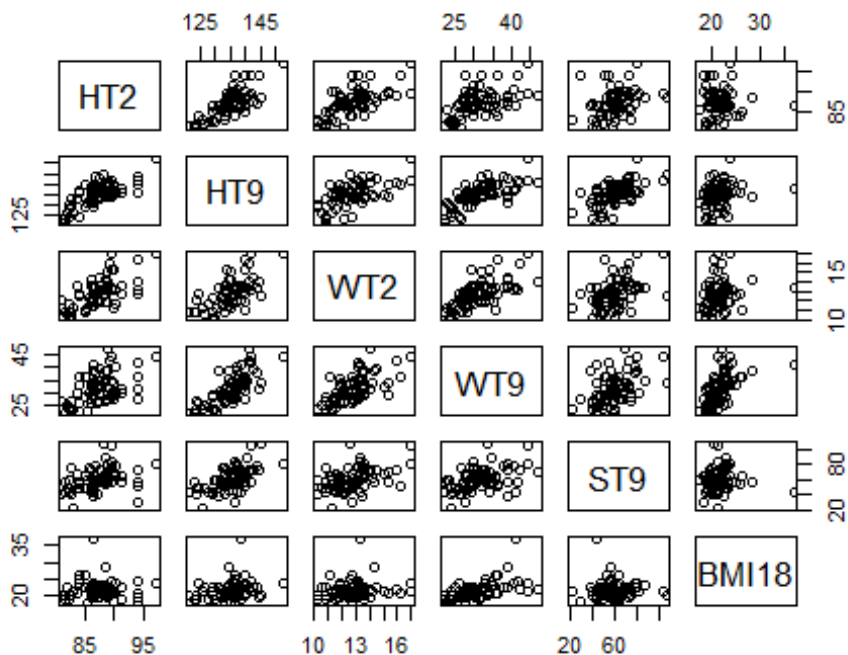
```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.4731 -1.4740  0.0786  1.9299  4.9411
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  25.9511    16.9685   1.529   0.142
## fall         0.1161     0.3341   0.348   0.732
##
## Residual standard error: 2.636 on 20 degrees of freedom
## Multiple R-squared:  0.006004,    Adjusted R-squared:  -0.0437
## F-statistic: 0.1208 on 1 and 20 DF,  p-value: 0.7318
```

on the two timelines approximately the values of the summary are the same.

#problem 3.3

draw a scatterplot of selected variables

```
m5<-BGSgirls[,c(2,4,1,3,6,11)]
pairs(m5)
```



finding correlation

```
cor(m5)
```

	HT2	HT9	WT2	WT9	ST9	BMI18
HT2	1.00000000	0.7383562	0.6445495	0.5229277	0.361724146	0.042573733
HT9	0.73835617	1.0000000	0.6071247	0.7276123	0.603368147	0.236907969
WT2	0.64454954	0.6071247	1.0000000	0.6925390	0.451581158	0.190947873

```
## WT9    0.52292768 0.7276123 0.6925390 1.0000000 0.453004062 0.545925753
## ST9    0.36172415 0.6033681 0.4515812 0.4530041 1.000000000 0.005603061
## BMI18 0.04257373 0.2369080 0.1909479 0.5459258 0.005603061 1.000000000
```

both the scatterplot and the values show that the variables are positively correlated

#3.3.3

for the given mean function

```
m6<-lm(BMI18~HT2+WT2+HT9+WT9+ST9,BGSgirls)
summary(m6)
```

```
##
## Call:
## lm(formula = BMI18 ~ HT2 + WT2 + HT9 + WT9 + ST9, data = BGSgirls)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -5.0948 -1.2186 -0.2533  1.0090 10.4951
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 30.855335   8.781156   3.514 0.000817 ***
## HT2         -0.193997   0.130819  -1.483 0.142996
## WT2         -0.317779   0.278736  -1.140 0.258505
## HT9          0.008057   0.096344   0.084 0.933613
## WT9          0.419762   0.075211   5.581 5.2e-07 ***
## ST9         -0.044416   0.022219  -1.999 0.049853 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.14 on 64 degrees of freedom
## Multiple R-squared:  0.4431, Adjusted R-squared:  0.3996
## F-statistic: 10.19 on 5 and 64 DF, p-value: 3.294e-07
```

the value of sigma^ and multiple R^2 are 2.14 and 0.4431

#hypothesis test

```
qt(1-0.1/2 , 64)
```

```
## [1] 1.669013
```

*#WT9 and ST9 have a t value more than the quantile level which rejects
null hypothesis.*

